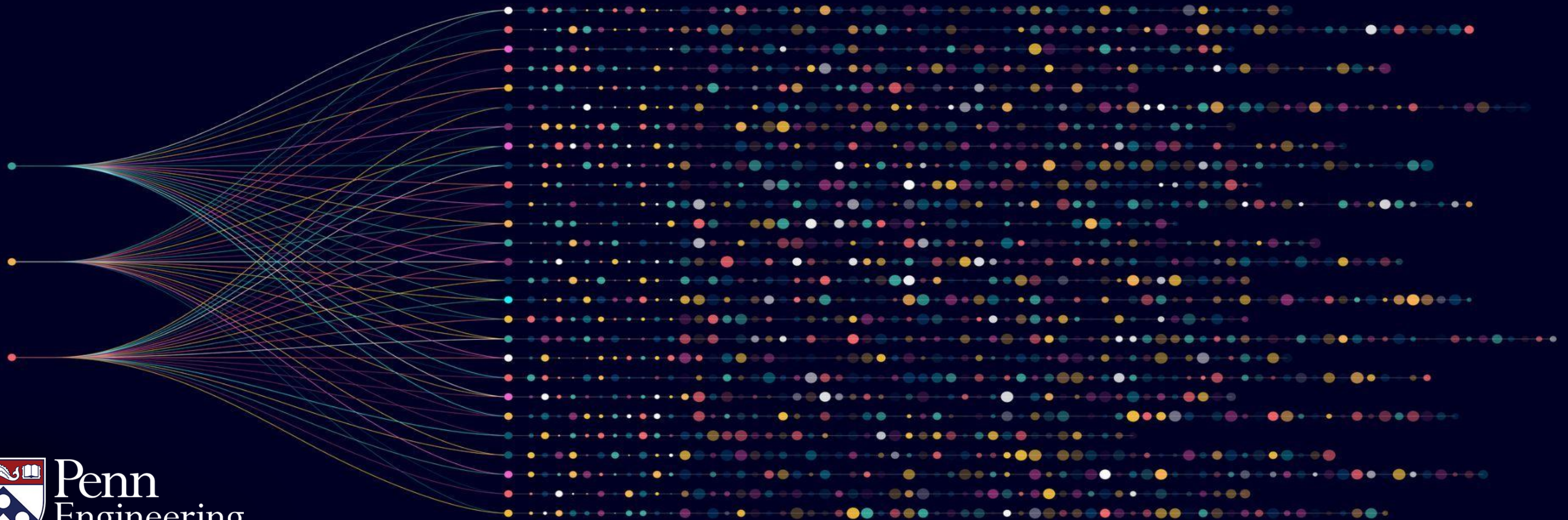


Algorithms for Big Data

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Markov's inequality



How big is the median household income?

- US Census Bureau: The per capita income in the US is ~ \$38,000

$$\frac{\text{Total annual income by everyone in the US}}{\text{US population}} \approx \$38,000.$$

Question: How big can the median income be?

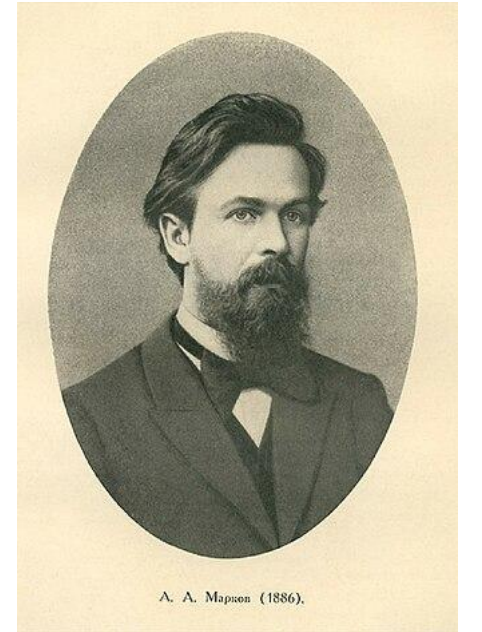
Is it possible that 50% or more people make \$80,000 or more?

Markov's inequality

Markov's inequality states that for any random variable $\mathbf{X}$ which is always non-negative and $t > 0$, $\Pr[\mathbf{X} \geq t] \leq \frac{E[\mathbf{X}]}{t}$

Corollary: if $\mathbf{X}$ is non-negative, then $\Pr[\mathbf{X} \geq 10E[\mathbf{X}]] \leq 1/10$

Markov's inequality is fundamental in probability theory – does not require any assumptions beyond non-negativity of $\mathbf{X}$



Proof of Markov's inequality

Markov's inequality: If $\mathbf{X}$ which is always non-negative and $t > 0$, $\Pr[\mathbf{X} \geq t] \leq \frac{E[\mathbf{X}]}{t}$.

Proof: By definition, $E[\mathbf{X}] = \sum_x x \cdot \Pr[\mathbf{X} = x]$.

$$E[\mathbf{X}] = \sum_{x < t} x \cdot \Pr[\mathbf{X} = x] + \sum_{x \geq t} x \cdot \Pr[\mathbf{X} = x] \quad - (1)$$

Observe that $\sum_{x < t} x \cdot \Pr[\mathbf{X} = x] \geq 0$ – since $\mathbf{X}$ can only take non-negative values and $\Pr[\mathbf{X} = x]$ is always non-negative

Thus, $E[\mathbf{X}] \geq \sum_{x \geq t} x \cdot \Pr[\mathbf{X} = x]$.

Proof of Markov's inequality

Markov's inequality: If $\mathbf{X}$ which is always non-negative and $t > 0$, $\Pr[\mathbf{X} \geq t] \leq \frac{E[\mathbf{X}]}{t}$.

Proof (continued):

So, fat we have $E[\mathbf{X}] \geq \sum_{x \geq t} x \cdot \Pr[\mathbf{X} = x]$ – (2)

$\sum_{x \geq t} x \cdot \Pr[\mathbf{X} = x] \geq \sum_{x \geq t} t \cdot \Pr[\mathbf{X} = x] = t \cdot (\sum_{x \geq t} \Pr[\mathbf{X} = x])$ – (3).

However, $(\sum_{x \geq t} \Pr[\mathbf{X} = x]) = \Pr[\mathbf{X} \geq t]$. Plugging into (3), $\sum_{x \geq t} x \cdot \Pr[\mathbf{X} = x] \geq t \cdot \Pr[\mathbf{X} \geq t]$.

Plugging back into (2), we get $E[\mathbf{X}] \geq t \cdot \Pr[\mathbf{X} \geq t]$ (same as our claim)

Application of Markov's inequality

Application #1: Relating Average and Median income

Let the population of a country be N and the incomes be $x_1, \dots, x_N \geq 0$.

Let $\mathbf{Z}$ be the random variable defined as: for each $1 \leq i \leq N$, with probability $\frac{1}{N}$, $\mathbf{Z} = x_i$.

Note that $\mathbf{Z} \geq 0$ and $E[\mathbf{Z}] = P$ – per capita income

By Markov's inequality, $\Pr[\mathbf{Z} \geq 2P] \leq \frac{E[\mathbf{Z}]}{2P} = \frac{1}{2}$.

Note that if the median income of the country is M^* , then $\Pr[\mathbf{Z} \geq M^*] = \frac{1}{2}$.

Thus, the median income $M^* \leq 2P$.

So, if the US per capita income is \$38,000, the median income cannot exceed \$76,000

Application #2: Number of fixed points of a random permutation

Recall the setting: $A = \{1, 2, \dots, n\}$ and let $\mathbf{B}$ be a random permutation of A .

We are interested in the number of fixed points of $\mathbf{B}$ (denoted by $\text{FP}(\mathbf{B})$)

– i.e., positions i which are unchanged in $\mathbf{B}$

Previous video: We established $E[\text{FP}(\mathbf{B})] = 1$

We now notice that $\text{FP}(\mathbf{B})$ is a non-negative random variable

Thus, Markov's inequality implies that for $c \geq 1$, $\Pr[\text{FP}(\mathbf{B}) \geq c] \leq \frac{E[\text{FP}(\mathbf{B})]}{c} = \frac{1}{c}$.

Aside: Montmort [1700s] established that $\Pr[\text{FP}(\mathbf{B}) = c] \sim \frac{1}{e \cdot c!}$

Summary

Markov's inequality – extremely powerful tool in probability theory

Can be applied in the most general of situations – only requirement is that the random variable is non-negative

Other strong probability bounds often obtained by applying Markov's inequality on the “right random variable”